Supplementary Material

Frontier-Centered Local Map Completion for Autonomous Exploration

Selma Güzel Erkan Uslu

This document provides supplementary results referenced in the main paper, including qualitative model comparisons (Figs. S1–S2), UNet variant evaluation (Table S1), and dataset size sensitivity results (Table S2).

Dataset Split Details

The dataset was initially partitioned into train (81.4%), validation (8.6%), and test (10%) sets at the environment level, yielding 9,418 train, 999 validation, and 1,160 test patches. To construct the larger dataset (BDS) without additional data collection, 1,000 patches were moved from the original test set into training, resulting in 10,418 train patches. The remaining 160 test patches form the shared test set used for evaluation of both SDS and BDS configurations. The validation set is identical across both configurations.

Qualitative Model Comparisons

Figures S1 and S2 show qualitative predictions of all evaluated models on the SDS test set. $\texttt{UNet}_{\texttt{base}}$ preserves thin walls and topological continuity most consistently. GAN variants (Pix2Pix-hinge, Seg-GAN) produce occasional hallucinations near obstacle boundaries. ViT generates fragmented wall predictions due to coarse 16×16 patch tokenisation. CVAE fails to capture structural patterns under irregular frontier masks. LSTM and ImageGPT produce near-uniform or structurally incoherent outputs.

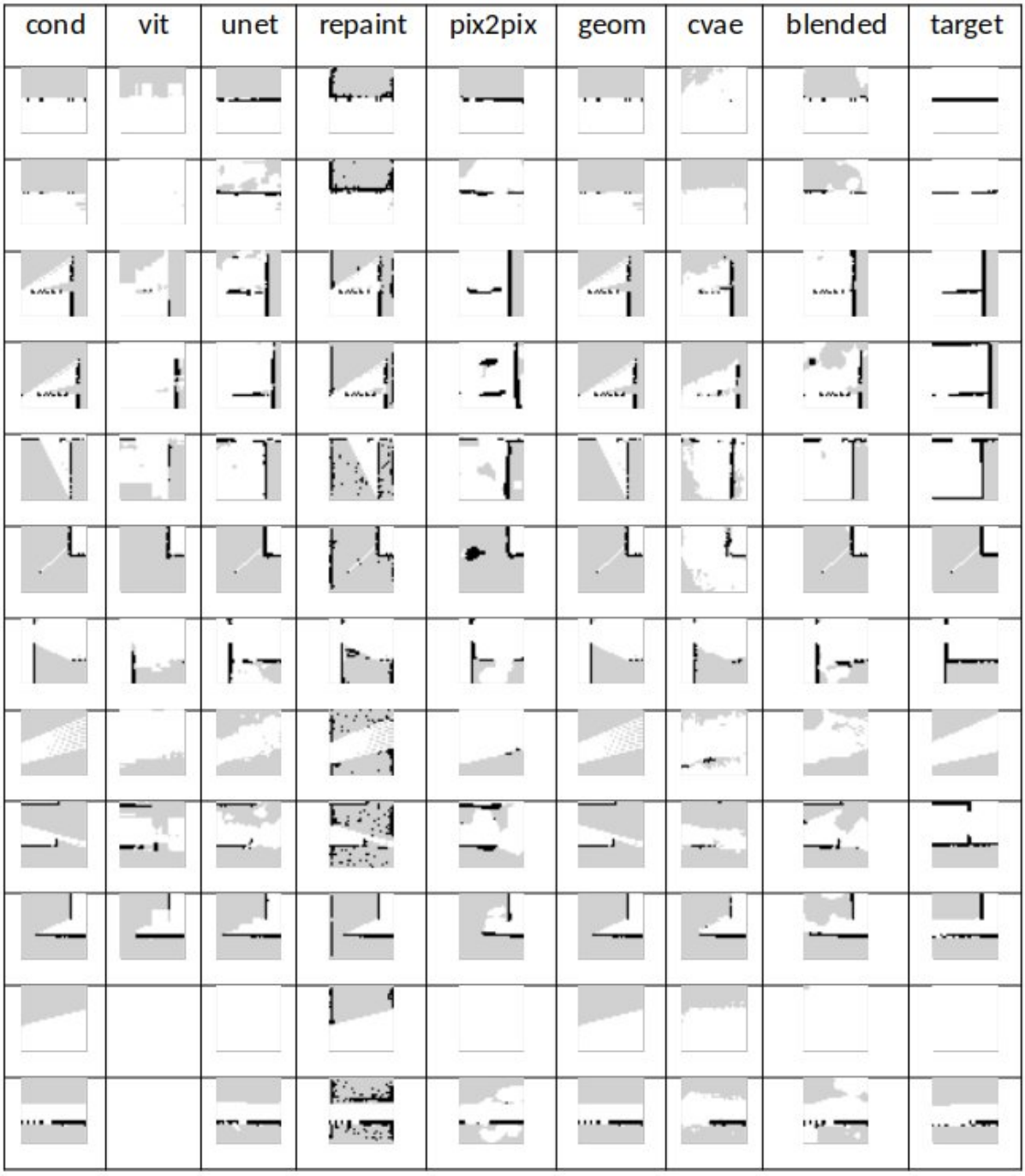


Figure S1: Qualitative comparison of frontier-centered completion models on SDS. Columns left to right: input condition, ViT, UNet, Pix2Pix, GEOM, CVAE, blended (known cells from input, predicted cells from model), target.

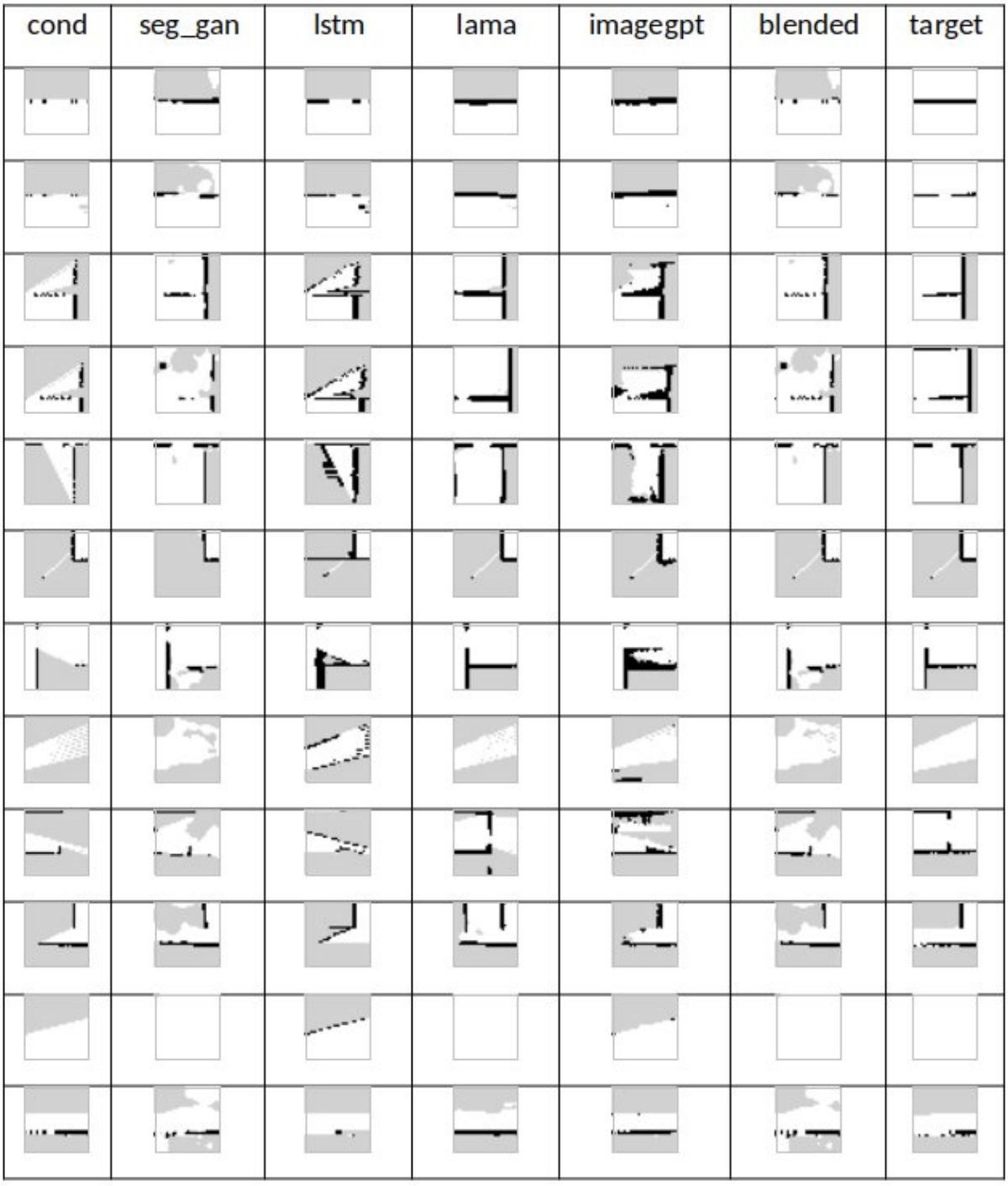


Figure S2: Qualitative comparison of frontier-centered completion models on SDS. Columns left to right: input condition, SegGAN, LSTM, ImageGPT, blended (known cells from input, predicted cells from model), target.

Table S1: UNet Variant Comparison (MASKED, SDS)

Table S1 reports MASKED evaluation results for all U-Net architectural variants on SDS. Since IoU and F1 differences are small (IoU range: 0.316–0.334), model selection is based on the average of all metrics (Pixel Acc, F1, IoU, L1, normalised throughput). $\texttt{UNet}_{\texttt{base}}$ achieves the highest overall average (0.627), driven by its superior Pixel Accuracy (0.649) and L1 (0.192), despite slightly lower IoU than deeper variants. LaMa, while a Fourier-augmented U-Net variant, is excluded from online deployment due to its throughput (12.67 img/s).

Table S1: UNet Variant Comparison: MASKED Evaluation on SDS. Selection criterion: average of all metrics (Pixel Acc, F1, IoU, L1, normalised throughput). † = excluded (throughput < 40 img/s).

Model	**Pixel Acc↑**	**F1-mac↑**	**IoU-mac↑**	**L1↓**	**img/s↑**
$\text{UNet}_{\text{base}}$	**0.649**	0.412	0.316	**0.192**	42.25
UNet_{new}	0.589	0.436	0.329	0.231	**44.47**
UNet_{att}	0.587	**0.441**	0.333	0.228	39.33
UNet (4 layers)	0.562	**0.441**	**0.334**	0.237	43.63
LaMa†	0.541	0.428	0.318	0.250	12.67

Table S2: Effect of Dataset Size (SDS → BDS)

Table S2 reports MASKED results when models are retrained on the larger dataset (BDS: 10,418 training patches vs. SDS: 9,418). $\texttt{UNet}_{\texttt{base}}$ performance remains nearly unchanged (IoU 0.309 vs. 0.316 on SDS), confirming the geometry-dominant nature of frontier-centered completion. GAN variants benefit modestly; ViT improves but remains below U-Net. This stability may not hold for substantially larger datasets, where data-hungry architectures such as ViT could potentially close the gap. RePaint and GEOM are omitted as they fall below the 40 img/s threshold or produce near-zero MASKED scores on SDS.

Table S2: MASKED Evaluation on BDS. U-Net performance is stable across dataset sizes; other families benefit more from additional data.

Model	**Pixel Acc↑**	**F1-mac↑**	**IoU-mac↑**	**L1↓**
$\mathbf{UNet_{base}}$	**0.619**	0.409	0.309	**0.206**
$\mathrm{UNet_{new}}$	0.566	**0.437**	**0.329**	0.238
Pix2Pix-hinge	≈0.60	≈0.39	≈0.30	≈0.21
ViT	0.570	0.323	0.242	0.236
CVAE	0.404	0.267	0.180	0.317